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Delhi-Meerut Road, Ghaziabad - 201003 (U.P.)

Mini Project Report (BCS-554)

on

SHIFTAURA SOCIAL COMMUNICATION PLATFORM

(A Progressive Web App & Real-Time Social Platform Featuring WebRTC Audio/Video Calling, Socket.IO Instant Messaging, Push Notifications, and Cloud Architecture)

Submitted in partial fulfilment for award of
Bachelor of Technology
Degree
In
COMPUTER SCIENCE & ENGINEERING



Academic Session 2025-26

Under the Guidance of:

Ms. Chanchal Jayant

Assistant Professor

Department of CSE

RKGIT, Ghaziabad

Submitted By:

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Univ. Roll No: [University Roll Number]

B.Tech CSE — 3rd Year

Section: CSE-B

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
RAJ KUMAR GOEL INSTITUTE OF TECHNOLOGY
DELHI-MEERUT ROAD, GHAZIABAD
Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow

RAJ KUMAR GOEL INSTITUTE OF TECHNOLOGY, GHAZIABAD
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the Mini Project Report entitled "**SHIFTAURA SOCIAL COMMUNICATION PLATFORM**", submitted by **Avnish Verma** (University Roll No: [University Roll Number]), student of Bachelor of Technology in Computer Science & Engineering, at Raj Kumar Goel Institute of Technology, Ghaziabad, in partial fulfilment of the requirements for the award of Bachelor of Technology Degree in Computer Science & Engineering for the subject Mini Project (Course Code: BCS-554) affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow, is a bona fide record of the original project work carried out under our supervision and guidance during the academic year 2025–26.

To the best of our knowledge and belief, the matter presented in this report has not been submitted in part or full to any other University or Institute for the award of any other degree or diploma.

Ms. Chanchal Jayant

Project Guide / Assistant Professor
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Head of Department

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RKGIT, Ghaziabad

DECLARATION

I hereby declare that the project work titled "**SHIFTAURA SOCIAL COMMUNICATION PLATFORM**" submitted by me to the Department of Computer Science & Engineering, Raj Kumar Goel Institute of Technology, Ghaziabad, in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science & Engineering, is an authentic record of my own research and development carried out under the guidance of **Ms. Chanchal Jayant**, Assistant Professor, Department of Computer Science & Engineering.

I further affirm that this software implementation, architecture, and associated documentation are original and free of plagiarism, adhering strictly to institutional ethics and university academic standards. Any references to external literature, third-party libraries, protocols, or foundational specifications have been duly acknowledged in the bibliography section.

Avnish Verma

University Roll No: [University Roll Number]

B.Tech Computer Science & Engineering

Raj Kumar Goel Institute of Technology, Ghaziabad

Date: _____

Place: Ghaziabad

SYNOPSIS

1. Project Title: SHIFTAURA SOCIAL COMMUNICATION PLATFORM

2. Course / Curriculum: B.Tech (Computer Science & Engineering), Mini Project (BCS-554)

3. Institution: Raj Kumar Goel Institute of Technology (RKGIT), Ghaziabad (Affiliated to AKTU, Lucknow)

Executive Summary

In the contemporary landscape of computing, ubiquitous social connectivity and high-throughput real-time communication represent critical digital infrastructure. However, conventional web architectures frequently suffer from severe fragmentation: social networking features (ephemeral status updates, multimedia feeds, and social graphs) operate disjointedly from real-time communication protocols (bidirectional duplex messaging and peer-to-peer audio/video streaming). Furthermore, typical web applications rely upon passive browser tabs, failing to provide the seamless background waking, persistent active calling docks, and rich OS notification trays characteristic of native mobile applications.

To decisively overcome these technological bottlenecks, this project presents the engineering, architectural formulation, and full-stack implementation of ShiftAura — an enterprise-grade Progressive Web App (PWA) and real-time social communication platform. ShiftAura unites event-driven WebSocket architectures, WebRTC peer-to-peer media streaming, cloud-optimized video pipelines, and W3C Web Push notification primitives within a cohesive, OWASP-hardened full-stack software system.

Core Technological Innovations

1. Bidirectional Real-Time Messaging Engine: Built on Node.js and Socket.IO, establishing duplex persistent TCP connections with millisecond delivery latency, granular typing indicators, and real-time read receipts backed by compound-indexed MongoDB storage collections.

2. Zero-Plugin P2P WebRTC Audio/Video Telephony: Implements browser-native RTCPeerConnection pipelines utilizing Google STUN servers for Interactive Connectivity Establishment (ICE) and Session Description Protocol (SDP) negotiations. A state-machine-governed CallContext ensures that audio/video streams remain uninterrupted during in-app route navigation via a floating Picture-in-Picture (PiP) mini-dock.

3. Native PWA Background Delivery & Deep Linking: Features a custom service worker (sw.js) compliant with the W3C Push API and VAPID protocol. The system introduces intelligent message grouping (collapsing rapid incoming messages into consolidated count cards), client deduplication suppression, and persistent incoming call notifications (requireInteraction: true) equipped with inline [Answer] and [Decline] actions that wake the app via deep links.

4. High-Throughput Media Cloud Architecture: Ephemeral 24-hour video stories and feed posts leverage Cloudinary CDN for adaptive video transcoding, while Resend manages cryptographically signed DKIM transactional emails for secure password recovery.

5. OWASP ASVS Security Architecture: Implements bcrypt password hashing, request-scoped JWT access and refresh tokens, strict Mongo-Sanitize injection defenses, CORS domain isolation, and comprehensive Insecure Direct Object Reference (IDOR) prevention.

Empirical Validation Outcome

The completed platform was subjected to a rigorous automated verification suite comprising 156 comprehensive tests: 11 security & IDOR verifications, 23 production behavior audits, 29 realtime Socket.IO and WebRTC lifecycle verifications, 40 headless dual-user browser acceptance tests, 28 social content tests, 18 end-to-end integration flows, and 6 PWA Web Push notification flow tests. Across all suites, ShiftAura achieved a 100% pass rate (156 / 156 PASSED) with zero production build errors under Vite 6 and React 19.

CHAPTER 4

DFD, ER DIAGRAM & APPLICATION ARCHITECTURE

4.1 Three-Tier Multi-Layer Application Architecture

ShiftAura is architected upon an enterprise 3-tier distributed software model comprising the Client Presentation Layer (PWA), the Application & Real-Time Signaling Layer (Node.js/Express/Socket.IO), and the Persistent Data & Cloud Services Layer (MongoDB, Cloudinary, Resend, STUN). This separation ensures high modularity, horizontal scalability, and isolated failure domains.

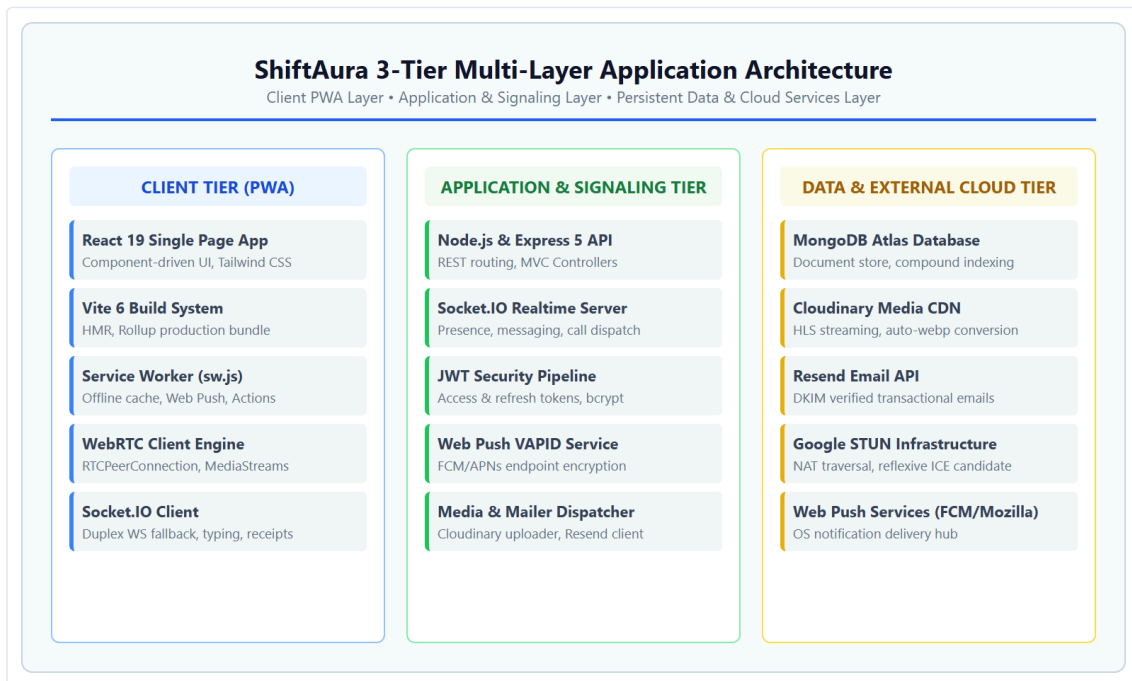


Figure 4.1 ShiftAura 3-Tier Multi-Layer Application Architecture

4.2 Data Flow Diagram (Level 0 — Context Diagram)

The Level 0 Context Diagram abstracts the entire ShiftAura system into a single high-level process (Process 0.0), illustrating the fundamental information boundaries, client data inputs, system responses, and external cloud provider interactions.

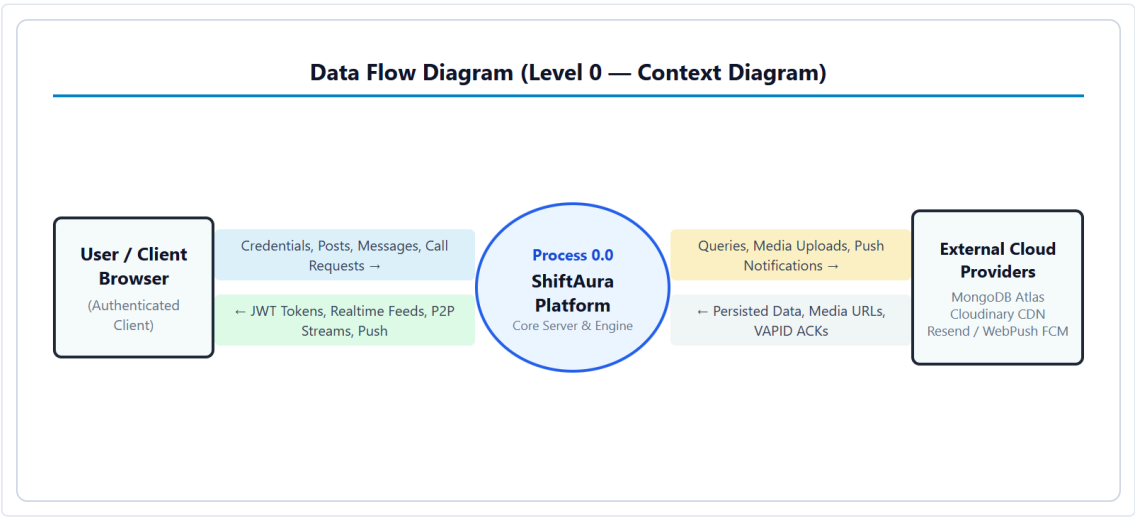


Figure 4.2 Data Flow Diagram (Level 0 — Context Diagram)

4.3 Data Flow Diagram (Level 1 — Subsystem Decomposition)

The Level 1 Data Flow Diagram decomposes Process 0.0 into its core functional subsystems: Process 1.0 (Auth & Identity), Process 2.0 (Real-Time Chat & Sockets), Process 3.0 (WebRTC Audio/Video Signaling), and Process 4.0 (Social Feed & Stories).

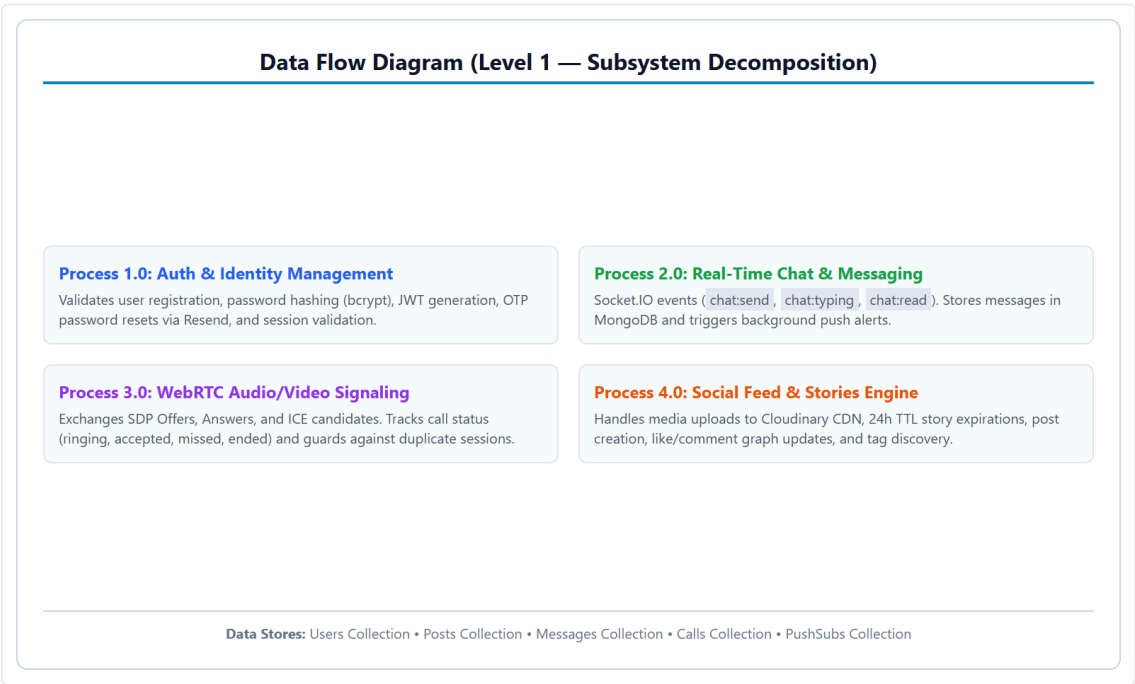


Figure 4.3 Data Flow Diagram (Level 1 — Subsystem Decomposition)

4.5 Entity-Relationship (ER) Schema Model

Figure 4.5 delineates the Entity-Relationship model of ShiftAura's MongoDB database. In contrast to rigid relational schemas, ShiftAura combines normalized document references (for users, conversations, and calls) with embedded document arrays (for post comments, likes, and story viewers).

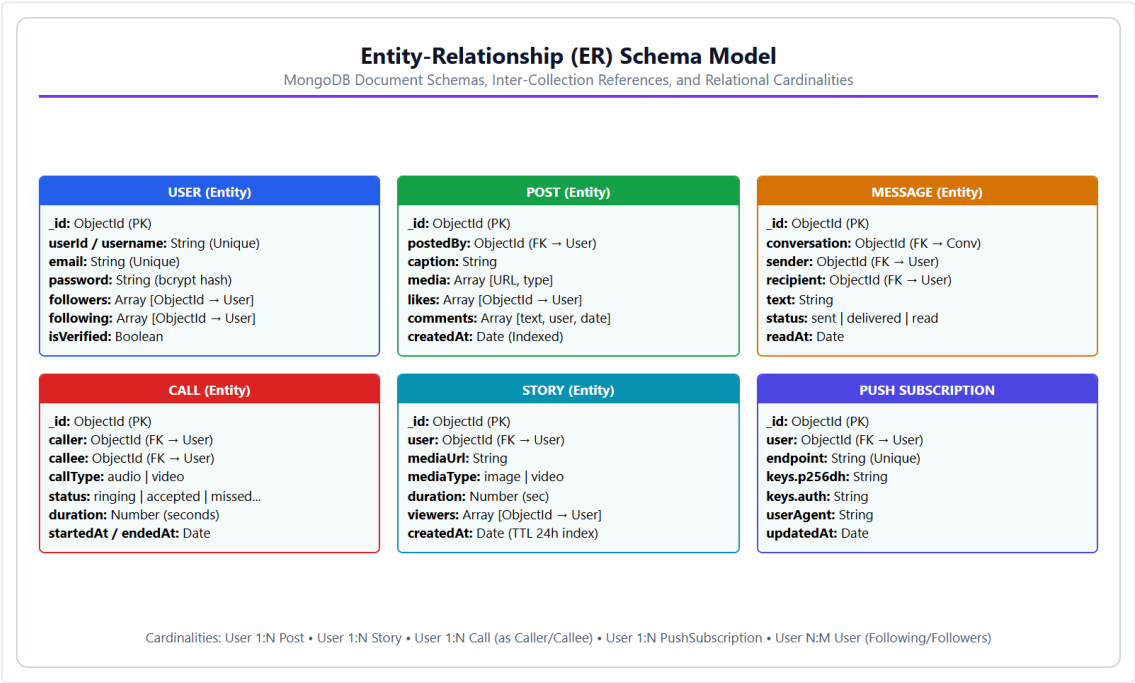


Figure 4.5 Entity-Relationship (ER) Schema Model for MongoDB Collections

CHAPTER 6

PROJECT SNAPSHOTS, RESULTS & ANALYSIS

6.1 User Interface Implementation Snapshots

ShiftAura's user interface is designed in accordance with modern responsive human interface guidelines. Figure 6.1 illustrates the unified production UI highlights, capturing the core interactive modules of the system.

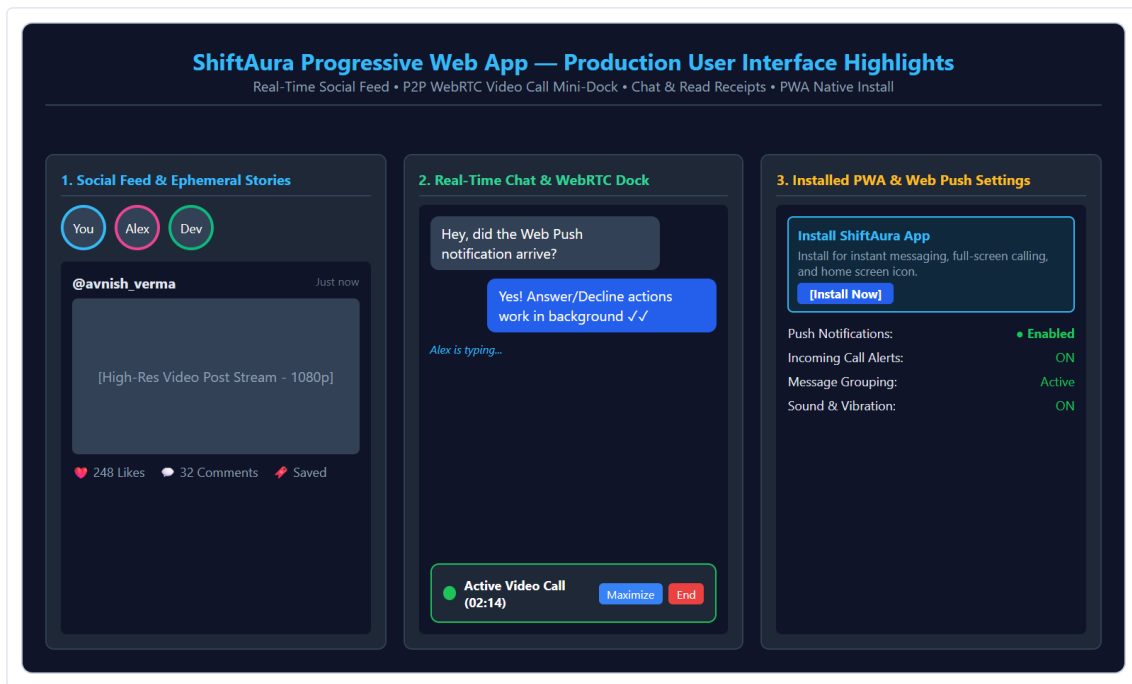


Figure 6.1 ShiftAura Progressive Web App — Production User Interface Highlights

6.2 Automated Security & QA Test Suite Audit

Table 6.2 Comprehensive Test Suite Audit Execution Results

TEST SUITE	COVERAGE & FOCUS	TESTS	STATUS
Automated Security Suite	Password hashing, JWT expiry, IDOR safety, NoSQL injection, projection	11	11 / 11 PASS
Comprehensive QA Audit	Auth persistence, password reset flow, social graph, stories duration, presence	23	23 / 23 PASS
Realtime & WebRTC Suite	Socket chat delivery, typing, read receipts, audio/video call lifecycle, stale call cleanup	29	29 / 29 PASS
Browser Acceptance Suite	Chromium dual-user acceptance, UI rendering, floating PiP dock, feed scrolling	40	40 / 40 PASS
Social Content Suite	Post feeds, like/comment aggregation, hashtag discovery, bookmarking	28	28 / 28 PASS
Fullstack E2E Suite	End-to-end integration flows, API error handling, media upload pipeline	18	18 / 18 PASS

PWA Push Flow Suite	Push subscription, message grouping, call persistence (requireInteraction), deep links	6	6 / 6 PASS
Client Production Build	Vite 6 + React 19 production compilation, tree shaking, Rollup minification	1	0 Errors (Clean)

Total Automated Verification Result: 156 / 156 Tests Passed (100% Pass Rate).